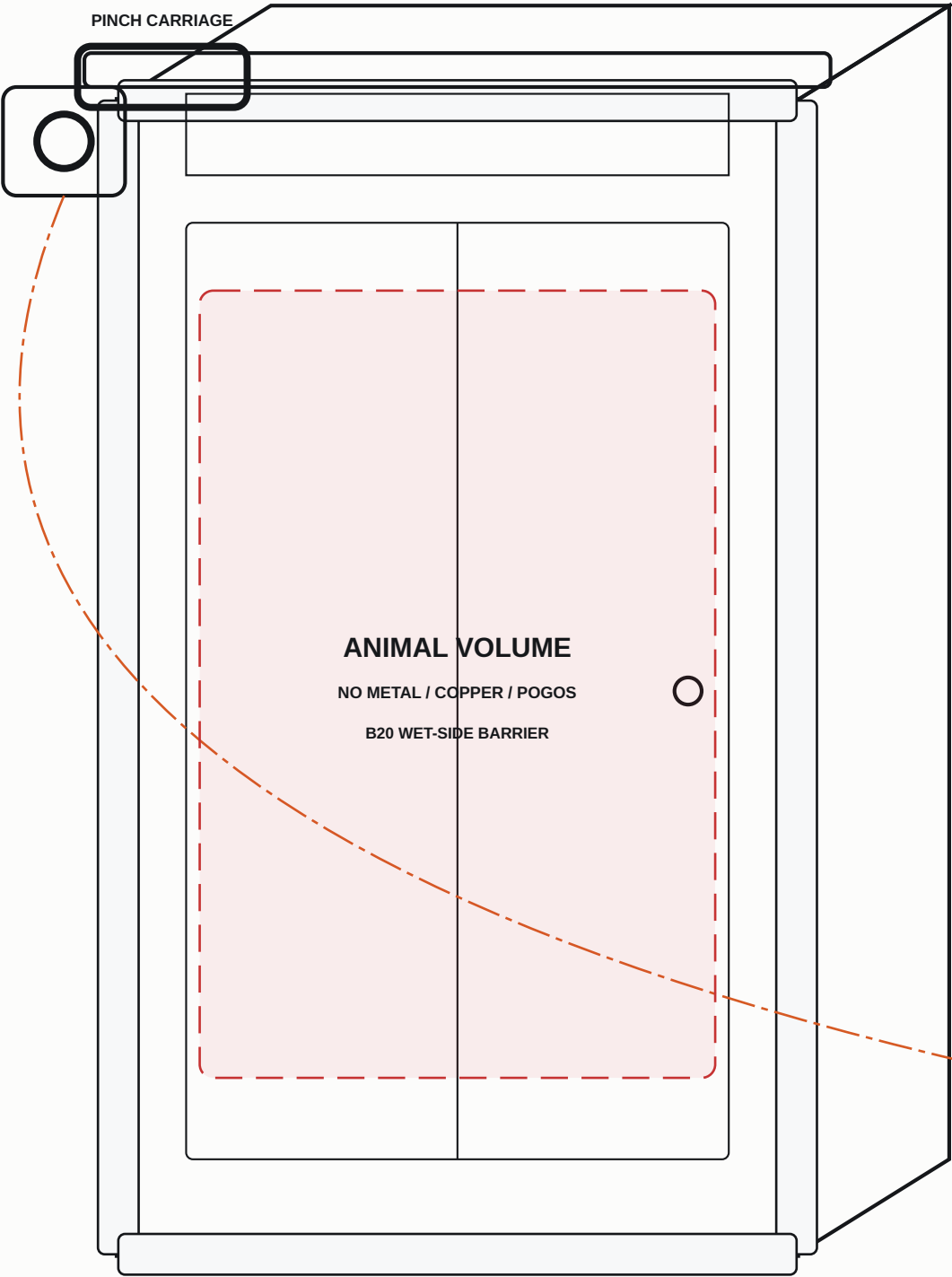


READABLE THEORY OF THE MACHINE

Exterior electrification; interior remains plastic, screened, and animal-safe.



SYSTEM THEORY

250 mm block perimeter; 500 mm first tower.
P01-P08: protected continuous VIN/GND/I2C/UID/P08.
P09-P12: indexed GMSL2 only while stationary.
Raw MIPI stays binder-local and Tachyon-local.
Pinch is break-before-move; video off while rolling.
M1/Tachyon brick never enters the animal cage.

LOCKED SAFETY BOUNDARY

Upper-third molt keep-out 142 mm CALCULATED.
Ceiling/vents: <= 0.8 mm non-metal LOCK.
B20 is the continuous wet-side barrier (CAD-01 bbox).
Door front/side; nymph-visible gap <=0.50 TARGET.
No taxon inferred: no live-mantis photo supplied.

MEASURED THIS RUN (DRAFT CAD/EE)

Base cursor/mantis-cad-02-carriage-9635
cdd523c55a63 CAD-02/#29
CAD-01/#28 fe8f875a80b3 PR 34
EE-23..26 PRs 37-40 (read-only)
No Particle camera SKU or CSI pin table drawn.

DRAWING STATUS

Views are NTS until released STEP drives scale.
OCCT bboxes from CAD-01 unique parts; kinematics from CAD-02.
S08 follows EE-24 net-map; KiCad is circuit authority.
docs/variant screenshots: none on PRs 32-40 this run.

NTS - DO NOT SCALE

NTS - DO NOT SCALE



PARTICLE-BASE MANTIS TERRARIUM

COVER / THEORY

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

SCALE
NTS

DATE
2026-08-20

REV
B-41

SHEET
01 OF 12 / S00

STATUS
THEORETICAL

DRAWN
GBG#41

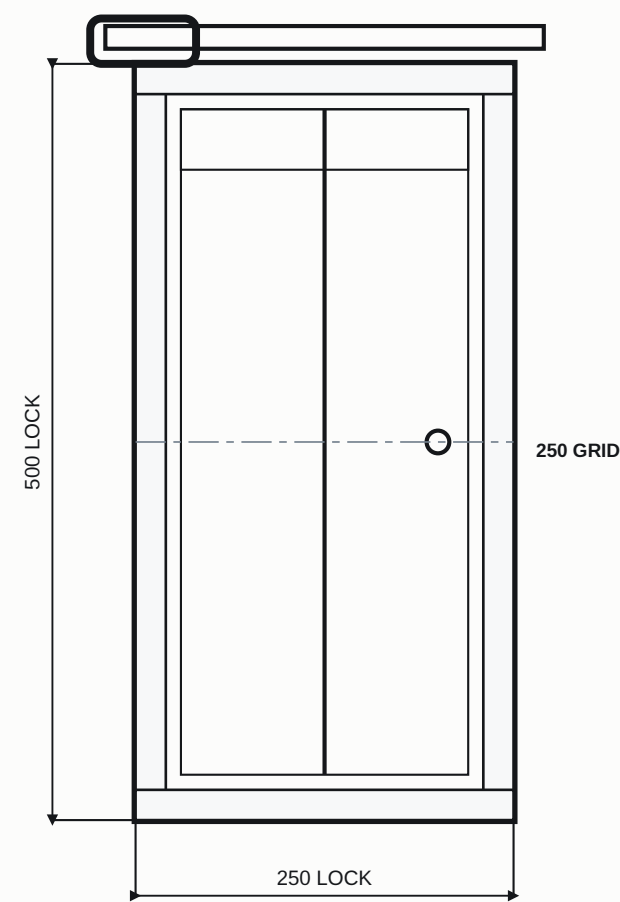
DWG
MT-S00

FORMAT
A3 SVG/PDF

TOL
AS NOTED

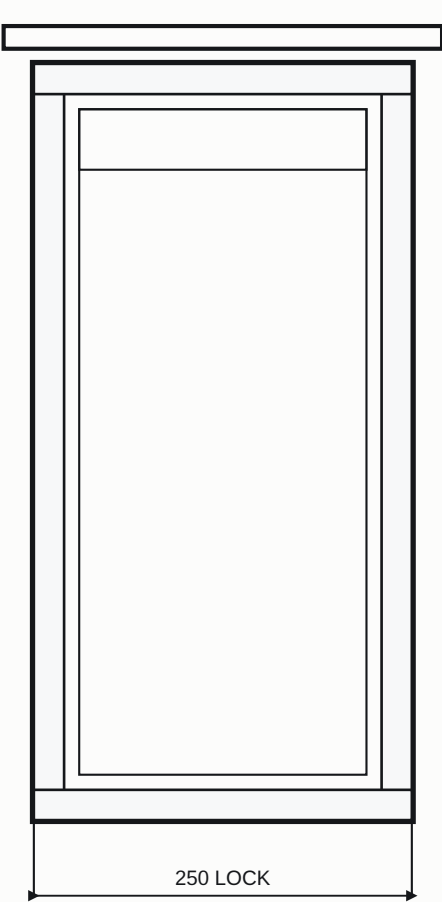
FRONT ELEVATION

SCALE NTS - DO NOT SCALE



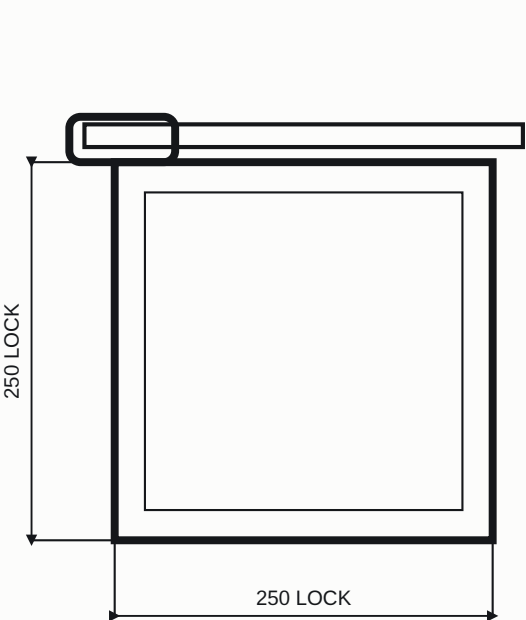
RIGHT ELEVATION

SCALE NTS - DO NOT SCALE



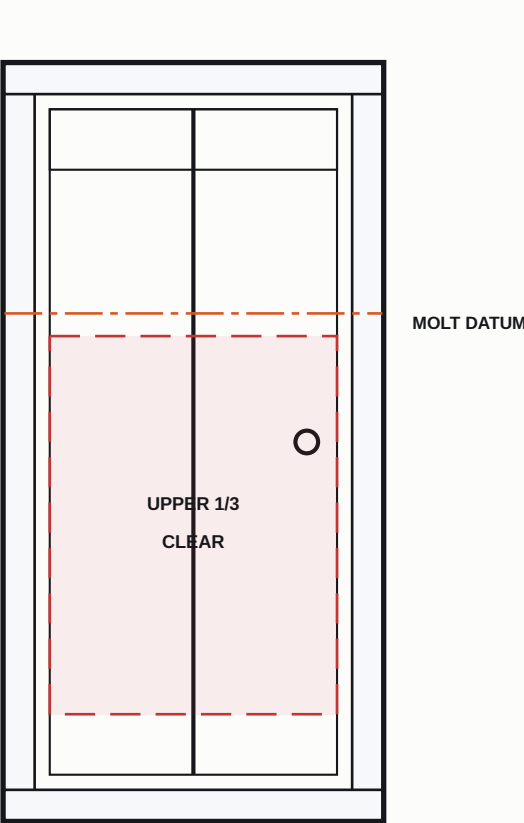
TOP PLAN

SCALE NTS - DO NOT SCALE



DATUM / KEEP-OUT

SCALE NTS - DO NOT SCALE



LOCKED ENVELOPE

Exterior: 250 W x 250 D x 500 H LOCK.
Module datum repeats at 250 mm.
Nominal clear: 202 x 202 x 427 CALCULATED.
CAD-01 assembly bbox: -43..258 X,
-43..250 Y, -8..508 Z REF.

PROJECTION

Third-angle arrangement shown.
Front is the door/view cassette face.
Rail offset 5 mm REF outside frame.
B20 starts at Y/X = -8 REF (offset+wall).

INTERFACE DATUMS

A: exterior block face (frame.world origin).
B: cassette seat plane.
C: rail conductor centerline (external).
D: false-bottom tray 25 mm REF.
Molt datum: 142 mm CALCULATED.

NTS - DO NOT SCALE

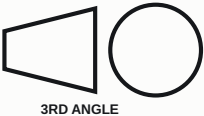
PARTICLE-BASE MANTIS TERRARIUM

ORTHOGRAPHIC ASSEMBLY

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

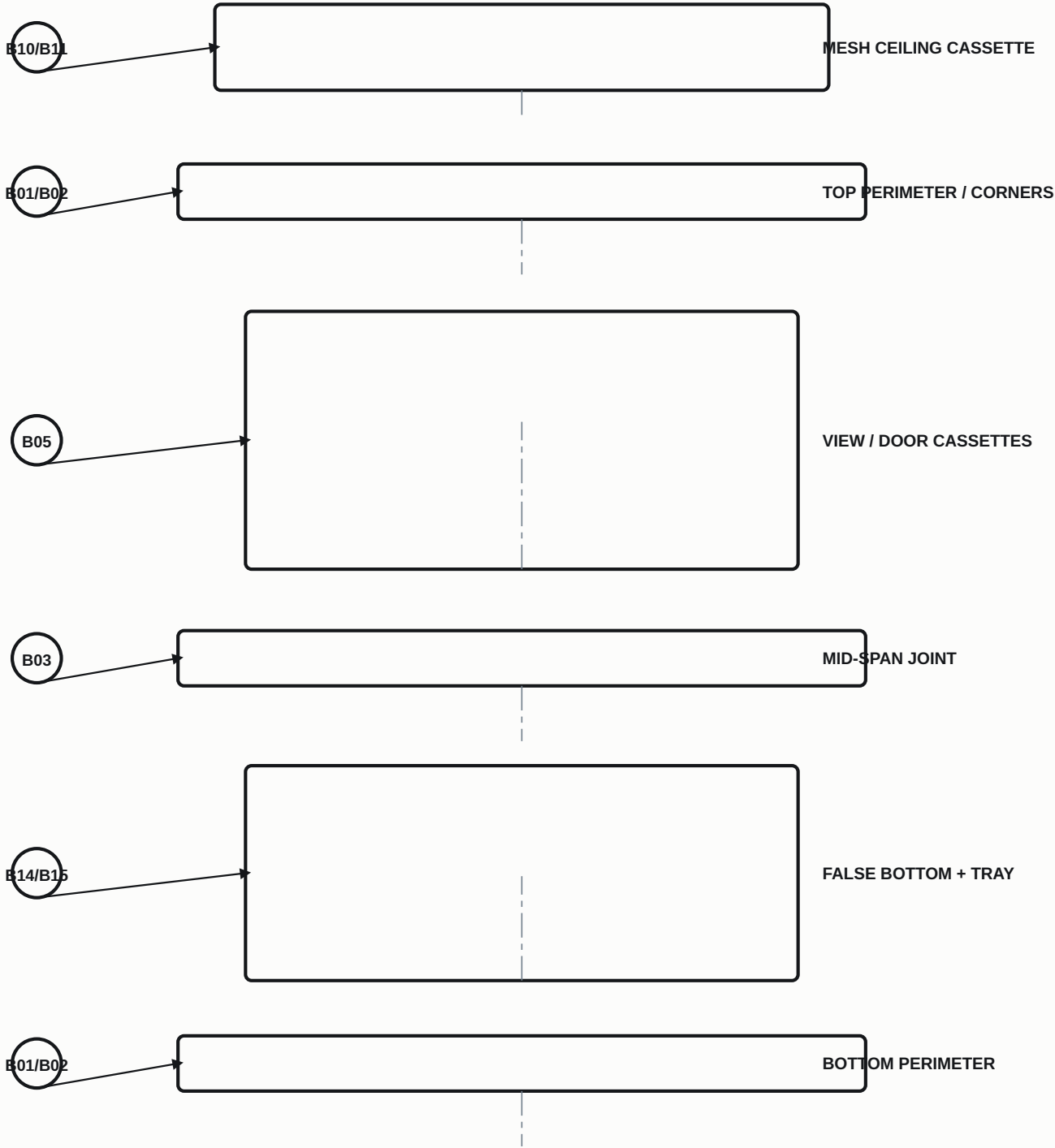
NTS - DO NOT SCALE



SCALE NTS	DATE 2026-08-20	REV B-41
SHEET 02 OF 12 / S01	STATUS THEORETICAL	DRAWN GBG#41
DWG MT-S01	FORMAT A3 SVG/PDF	TOL AS NOTED

FUNCTIONAL-AXIS EXPLOSION

SCALE NTS - DO NOT SCALE

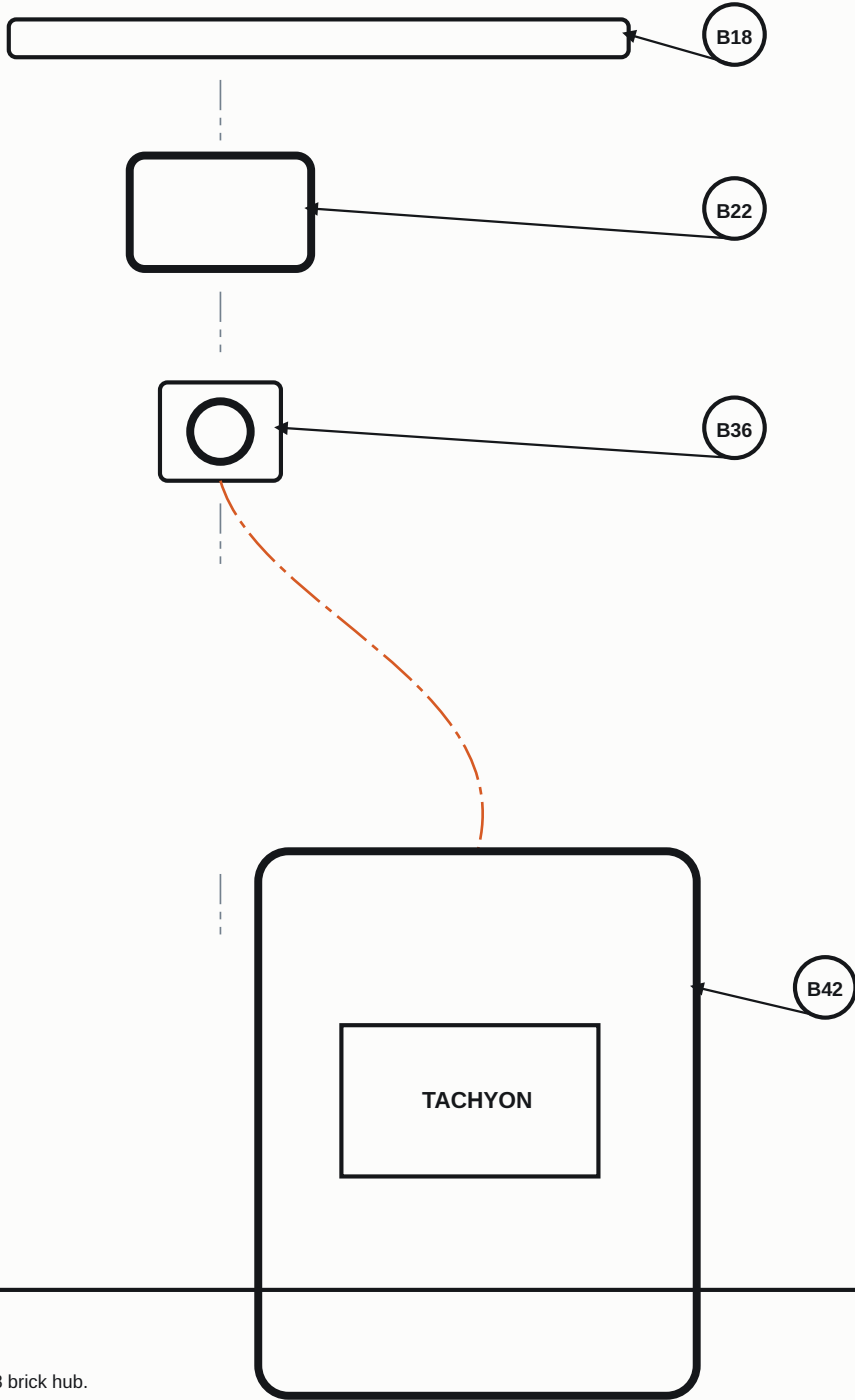


EXPLOSION RULE

Subassemblies remain grouped; trail lines follow service axes.
CAD-01 unique parts: B01-B04, B10, B18/B51/B52, B05-B07, B12-B14, B20.
CAD-02 grouped: B22-B27 pinch, B28/B29/B50 binder. No animal-side metal.

INSTALL ORDER

1 perimeter -> 2 faces -> 3 ceiling -> 4 false bottom.
5 animal-isolated rail -> 6 carriage -> 7 SerDes binder -> 8 brick hub.
First article requires leak, pinch, pull-off and nymph-gap checks.



PARTICLE-BASE MANTIS TERRARIUM
EXPLODED ASSEMBLY

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

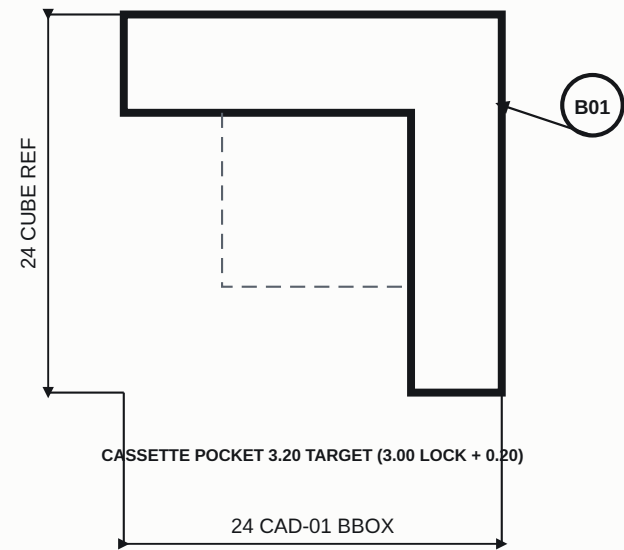
Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

NTS - DO NOT SCALE

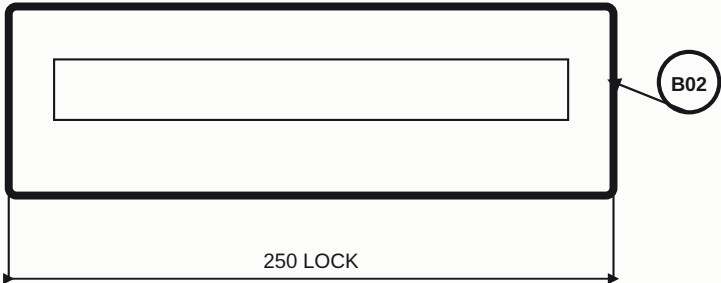


SCALE NTS	DATE 2026-08-20	REV B-41
SHEET 03 OF 12 / S02	STATUS THEORETICAL	DRAWN GBG#41
DWG MT-S02	FORMAT A3 SVG/PDF	TOL AS NOTED

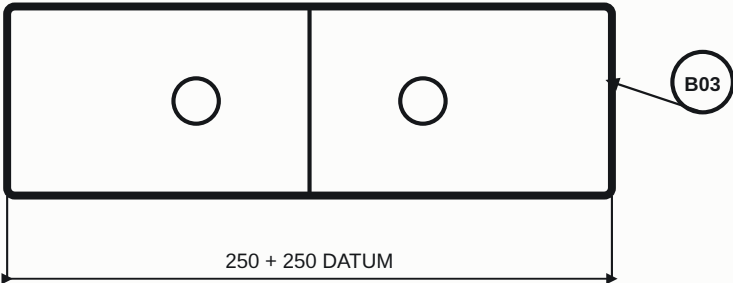
CORNER BLOCK
SCALE NTS - DO NOT SCALE



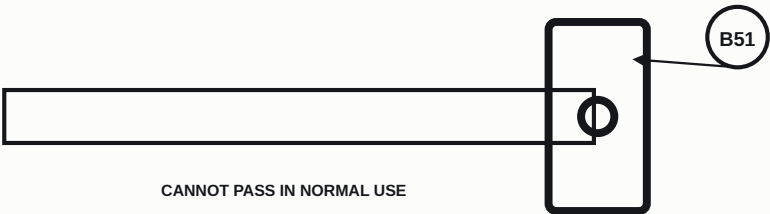
EDGE BLOCK
SCALE NTS - DO NOT SCALE



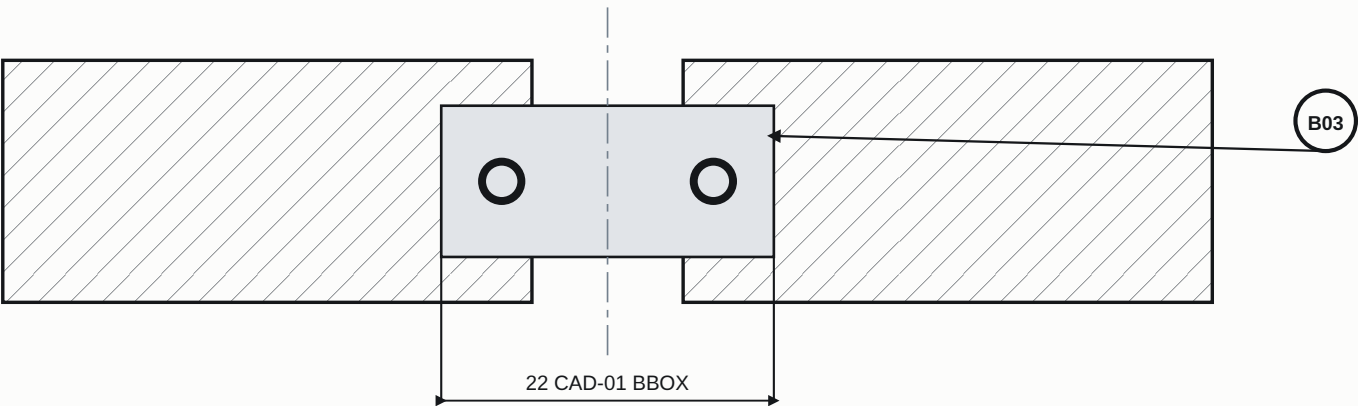
MID-SPAN BLOCK
SCALE NTS - DO NOT SCALE



ROUTE END STOP / SERVICE REMOVAL
SCALE NTS - DO NOT SCALE

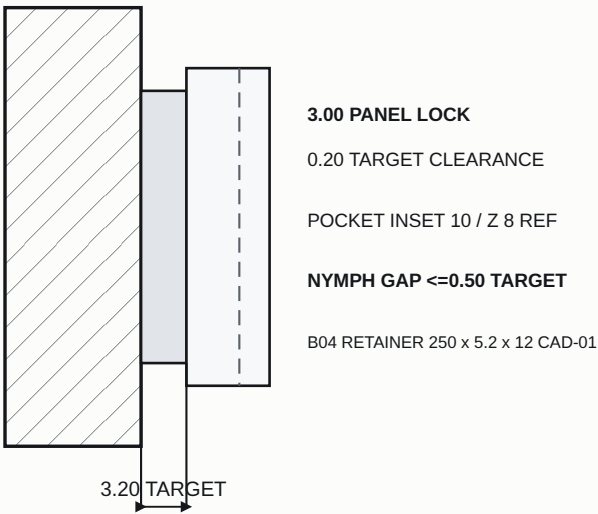


DETAIL A - 250 -> 500 SPAN JOINT
SCALE NTS - DO NOT SCALE



SHEAR BY TONGUE; B21 IS NOT A CORNER ROLL SOLID

DETAIL B - CASSETTE SEAT
SCALE NTS - DO NOT SCALE



CAD-01 MEASURED UNIQUE PARTS

B01 24 cube; B02 250 x 24 x 24; B03 22 x 24 x 24; B10 250 x 250 x 6.
Kerf 0.15 REF on B05/B06 cut profiles. Coupon 3.10/3.20/3.30.

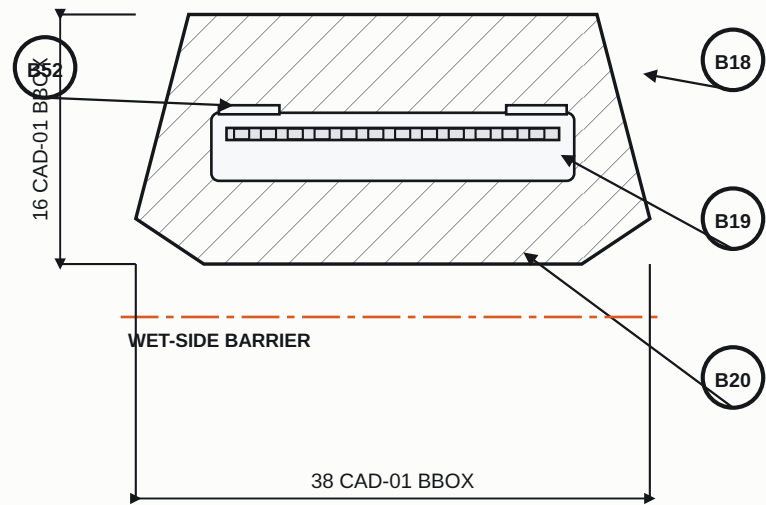
LOAD PATH / ROUTES

Panel -> gasket -> pocket wall -> perimeter block.
500 span bending crosses B03 coupler, not the face cassette.
B21: top-front and vertical routes are independent; no corner roll.
B51 8 x 38 x 17.5; bore 3.4 UNVERIFIED PN.

SCALE NTS	DATE 2026-08-20	REV B-41
SHEET 04 OF 12 / S03	STATUS THEORETICAL	DRAWN GBG#41
DWG MT-S03	FORMAT A3 SVG/PDF	TOL AS NOTED

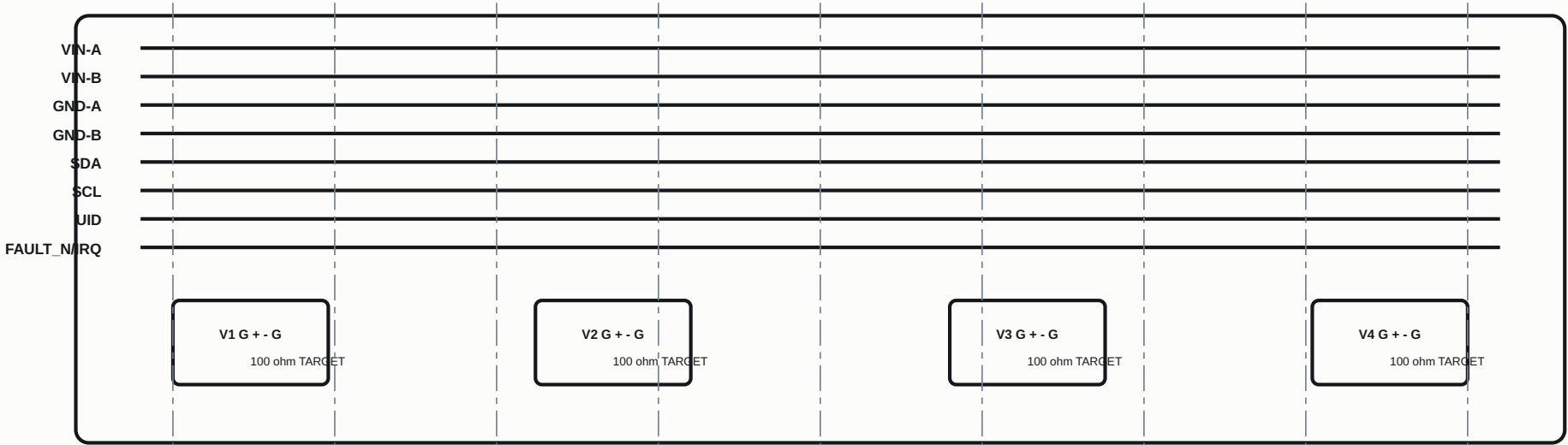
RAIL CHANNEL SECTION C-C

SCALE NTS - DO NOT SCALE



LONGITUDINAL CONDUCTOR RUN

SCALE NTS - DO NOT SCALE

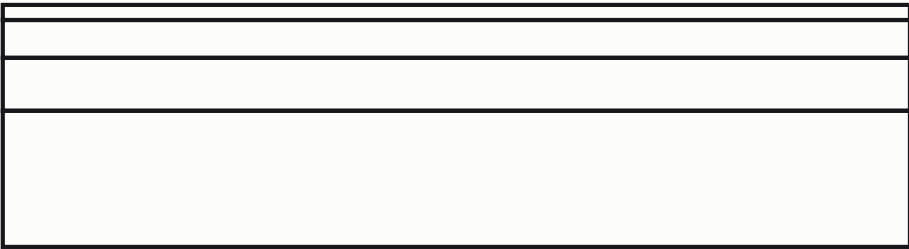


V-DOCK PITCH 125 CALCULATED (500/4); VIDEO ONLY AT INDEXED DOCKS

DETAIL D - STRIP STACK-UP

SCALE NTS - DO NOT SCALE

EXPOSED ENIG LAND IN CAPTIVE EXTERNAL SLOT - NO FILM OVER CONTACT



LAND t=0.2 REF (study); OTHER STACK LAYERS UNVERIFIED

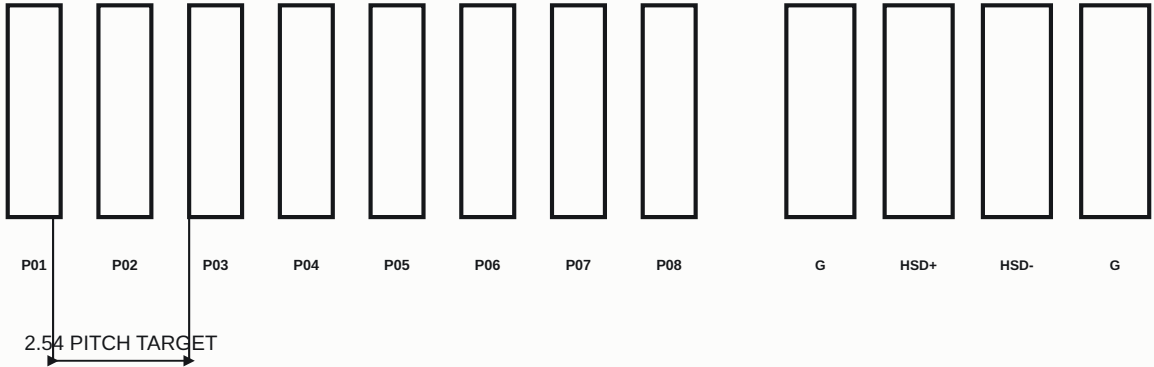
ENIG CONTACT LAND
FLEX / SPRING COPPER

DIELECTRIC

PETG / ASA FRAME

DETAIL E - CONTACT LANDS

SCALE NTS - DO NOT SCALE



FIELD 29.44 IN CAVITY 32; P09-P12 GMSL2 CELL

CAD-01 RAIL MEASURE

B18 250: 250 x 38 x 16; 500 span sibling.
Offset 5 REF; B52 lip 1.5; slot 0.4 REF.
Dock pad 18 REF; V-pitch 125 CALCULATED.
B20 is the wet-side barrier; it is not a film over contacts.

FAILURE CONTROL

S1 first travel opens Q1 before B27 lift (CAD-02 q=1.0 TARGET).
Bent HSD pogo -> link loss; no fallback to raw MIPI on the strip.
Each V-dock is point-to-point to one MAX96724 input.
B52 is a labyrinth wiper, not a hermetic cover. EE-23 series UNVERIFIED.

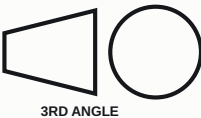
PARTICLE-BASE MANTIS TERRARIUM

RAIL + STRIP

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

NTS - DO NOT SCALE



3RD ANGLE

SCALE
NTS

SHEET
05 OF 12 / S04

DWG
MT-S04

DATE
2026-08-20

STATUS
THEORETICAL

FORMAT
A3 SVG/PDF

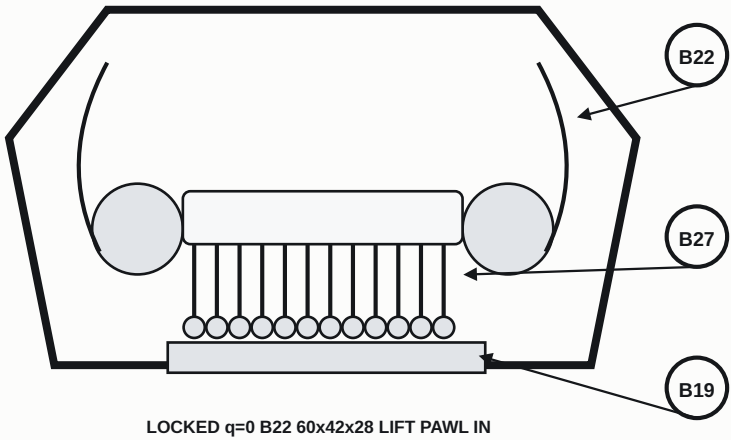
REV
B-41

DRAWN
GBG#41

TOL
AS NOTED

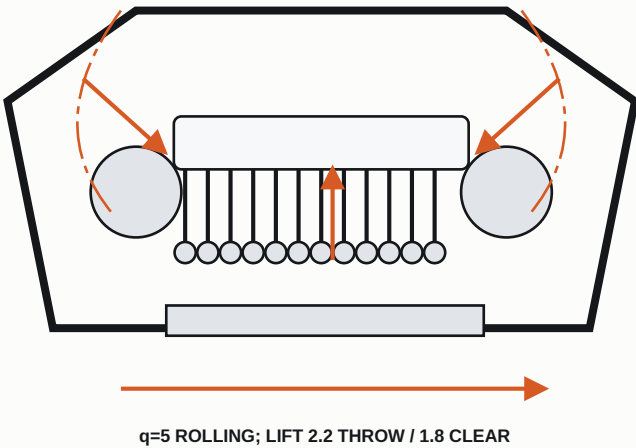
LOCKED STATE

SCALE NTS - DO NOT SCALE



PINCHED / TRAVEL STATE

SCALE NTS - DO NOT SCALE



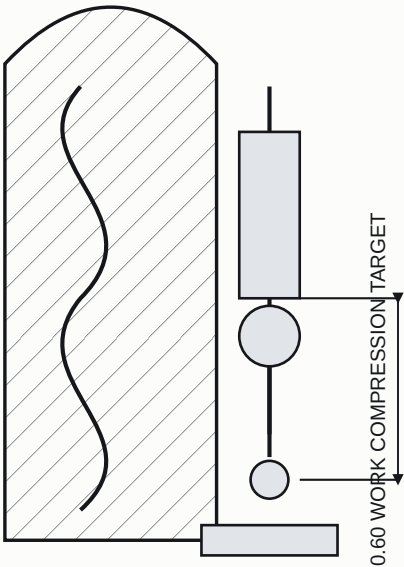
CAD-02 PINCH Q (MM)

SCALE NTS - DO NOT SCALE

q=0 LOCKED S1 closed; pawl in
q=1 S1_OPEN Q1 off; contacts seated
q=2.2 PINCH_SAFE lift pawl retracts
q=4 CLEAR lift={CAD02[throw]:g}; still locked
q=5 ROLLING translation pawl out

DETAIL F - PINCH / SPRING / POGO ARRAY

SCALE NTS - DO NOT SCALE



- B25 SPRING PN UNVERIFIED (POCKET ONLY)
- PINCH FORCE 15-25 N TARGET
- B27 PROXY d=1 / BARREL 6 UNVERIFIED
- B24 CARRIER 36x32x6; B26 d=8 UNVER
- MIPI IS NOT ON THESE CONTACTS
- ADR-002: HARDWARE PAWLS, NOT A TIMED DELAY

MECHANISM NOTE - PINCH-LIFT / ROLL

States: LOCKED / S1_OPEN / PINCH_SAFE / CONTACTS_CLEAR / ROLLING.
What moves: B24/B27 lift then rollers; rail/B19 stays grounded.
q=1 opens S1 envelope; contacts stay seated until q=2.2.
Lift pawl blocks Z until PINCH_SAFE. Throw 2.2 covers 1.8 clearance.
Translation pawl blocks roll until lift >= clearance. Speed cannot skip a pawl.
EE-24 S1/S2/Q1 parts UNVERIFIED; electrical dwell is a later independent gate.
Chosen over screws for one-hand relocate without opening the cage.

FAILURE / ACCEPTANCE

Interrupted pinch: keep Q1 off; require a clean full remate.
Bent HSD: training timeout; Q1 latches off. No MIPI fallback.
B51 retains loaded carriage (EE-26 route-retention UNVERIFIED).
No conductor may breach B20.

PARTICLE-BASE MANTIS TERRARIUM

CARRIAGE MECHANISM

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

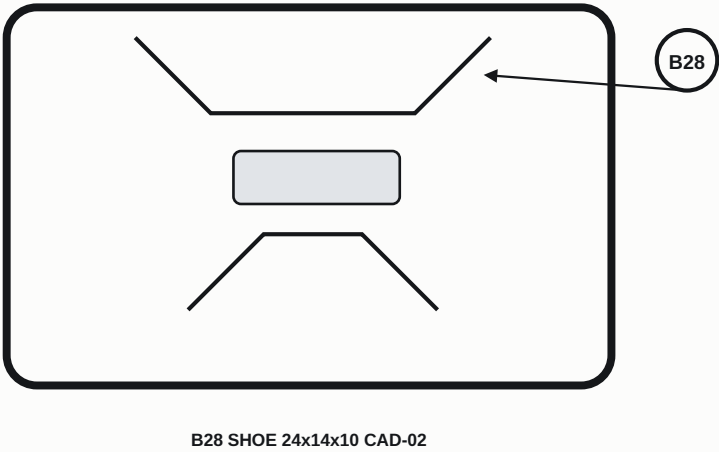
NTS - DO NOT SCALE



SCALE NTS	DATE 2026-08-20	REV B-41
SHEET 06 OF 12 / S05	STATUS THEORETICAL	DRAWN GBG#41
DWG MT-S05	FORMAT A3 SVG/PDF	TOL AS NOTED

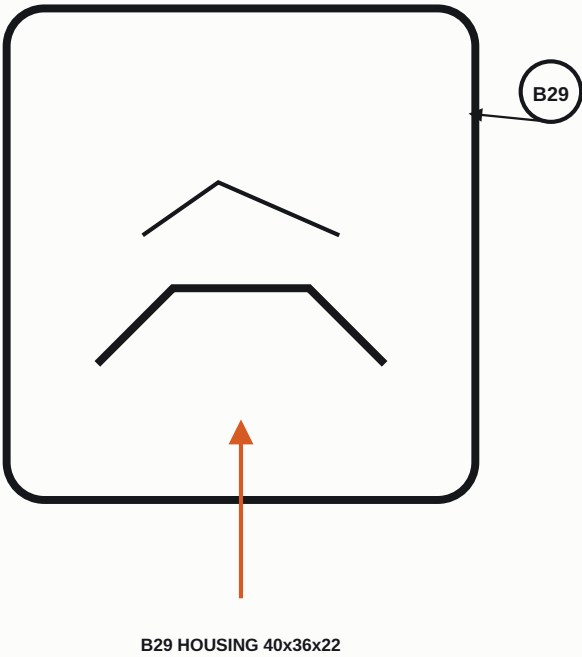
UNIVERSAL SHOE - FREE

SCALE NTS - DO NOT SCALE



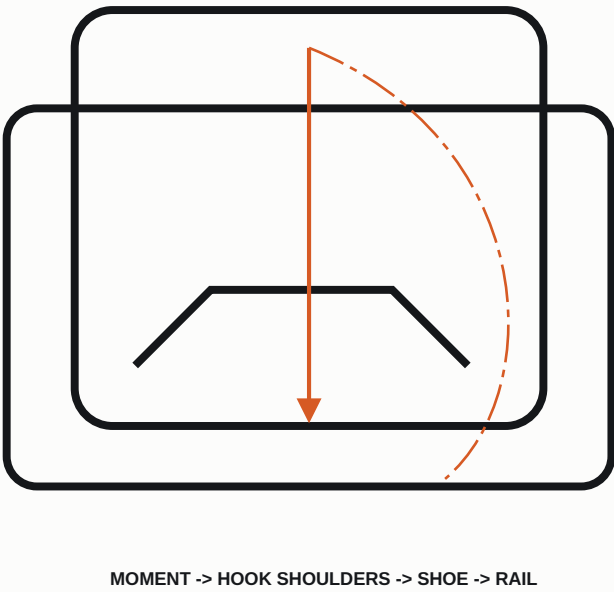
BINDER - MATING

SCALE NTS - DO NOT SCALE



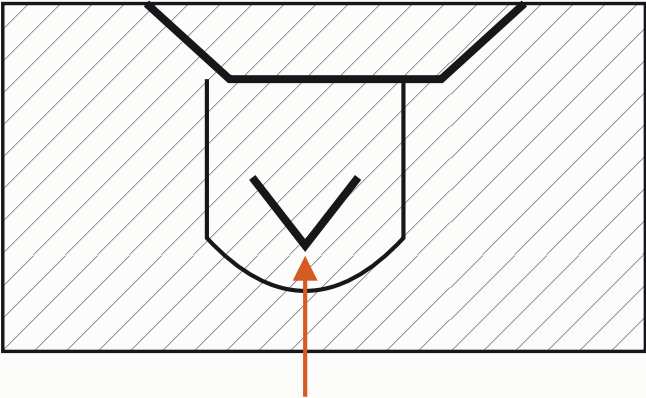
MATED / LOAD PATH

SCALE NTS - DO NOT SCALE



DETAIL G - CLICK LATCH SECTION

SCALE NTS - DO NOT SCALE



SPRING TONGUE DEFLECTION 1.2 TARGET

PULL-OFF ≥ 40 N TARGET

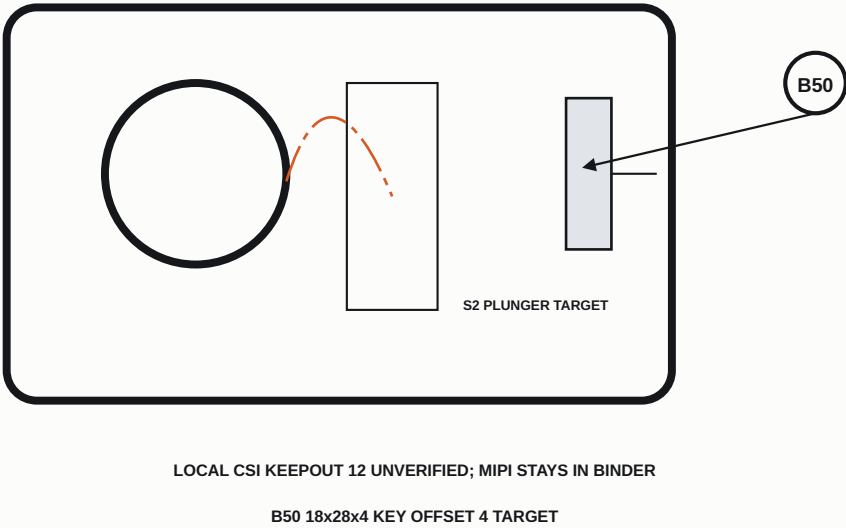
SIDE PLAY ≤ 0.20 TARGET

MATERIAL: PETG/ASA BODY + POM WEAR KEY REF

FAILURE: CRACK / PARTIAL ENGAGEMENT -> RETENTION WITNESS MARK

DETAIL H - BINDER-LOCAL FPC + B50 FACE

SCALE NTS - DO NOT SCALE



MECHANISM NOTE - BIND-LATCH

States: BLOCKED until pinch $q \geq 2.2$; then $r = 1/2/3.5$ TARGET.

Binder housing moves; B28 shoe stays on carriage. Load path bypasses B50.

First r opens S2 envelope while B50 stays keyed/seated (BRANCH_SAFE dwell).

Release pin retracts only at PINCH_SAFE. Firmware cannot move that pin.

FAILURE / CLEARANCE

Partial click: S2 stays open; Q1 remains off. Key blocks 180 reverse mate.

B50 series/pinout/SI launch UNVERIFIED (#23). Do not invent a connector.

Camera keep-out 32 is not a sourced module outline.

PARTICLE-BASE MANTIS TERRARIUM

UNIVERSAL LATCH + BINDER

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

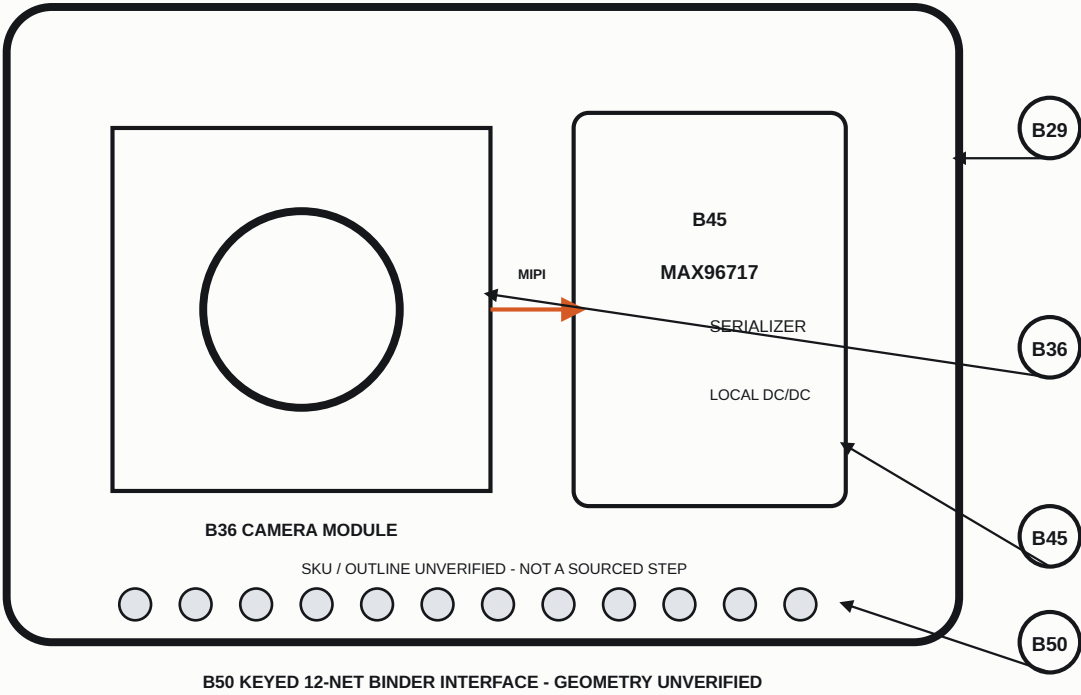
NTS - DO NOT SCALE



SCALE NTS	DATE 2026-08-20	REV B-41
SHEET 07 OF 12 / S06	STATUS THEORETICAL	DRAWN GBG#41
DWG MT-S06	FORMAT A3 SVG/PDF	TOL AS NOTED

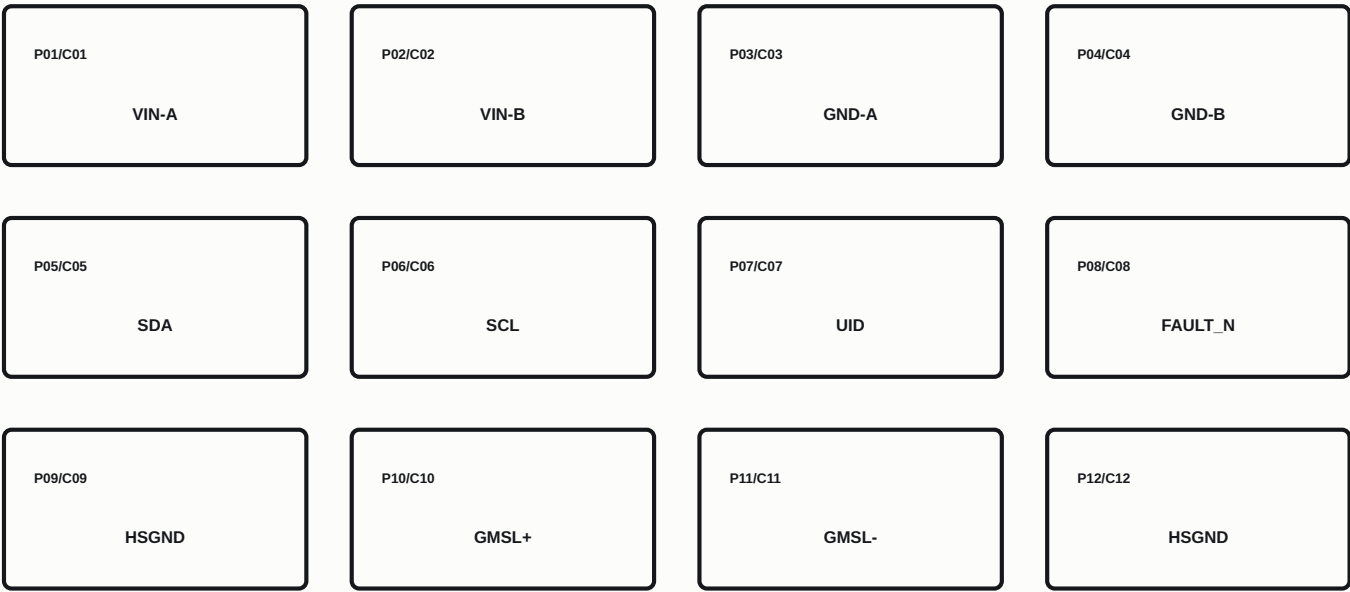
CAMERA BINDER ASSEMBLY

SCALE NTS - DO NOT SCALE



RAIL PXX / BINDER CXX MIRRORED NETS

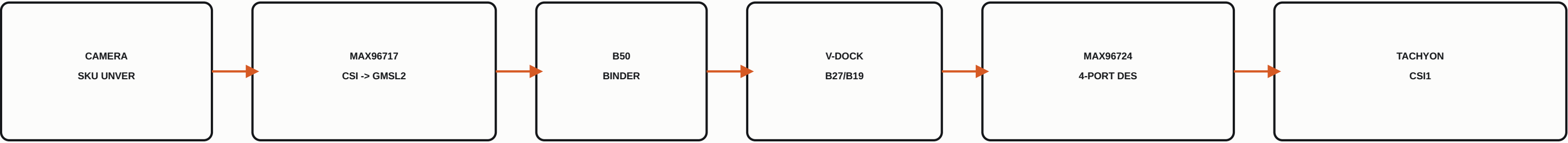
SCALE NTS - DO NOT SCALE



B27 + B50 FULL CHANNEL: 100 ohm TARGET / TO BE CHARACTERIZED

VIDEO PATH - NO TETHER TO MOVING CARRIAGE

SCALE NTS - DO NOT SCALE



RAW MIPI STAYS INSIDE BINDER; 3/6 Gbps GMSL2 CROSSES LOCKED POGO DOCK

POWER + VIDEO + REVERSE CONTROL CROSS THE CARRIAGE; NOTHING TETHERS IT

MECHANISM NOTE - CAMERA VIDEO DOCK

States: free / S1+S2 power-mated / training / link-trained / pinch-safe.
What moves: binder. V-dock and MAX96724 stay on rail/brick. No tether.
Raw MIPI never rides B50, B27, or the continuous strip (EE-23/25 lock).

FAILURE / WHY / EE-25

Partial dock or bent HSD: training timeout; Q1 latches off.
Particle lists IMX519 AF and S5K3P9SX sensors; no orderable SKU drawn.
EE-26 coupons: 2x-thru, B27-only, B50-only, full channel. Not animal-use.

PARTICLE-BASE MANTIS TERRARIUM

CAMERA LOAD / SERDES BINDER

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

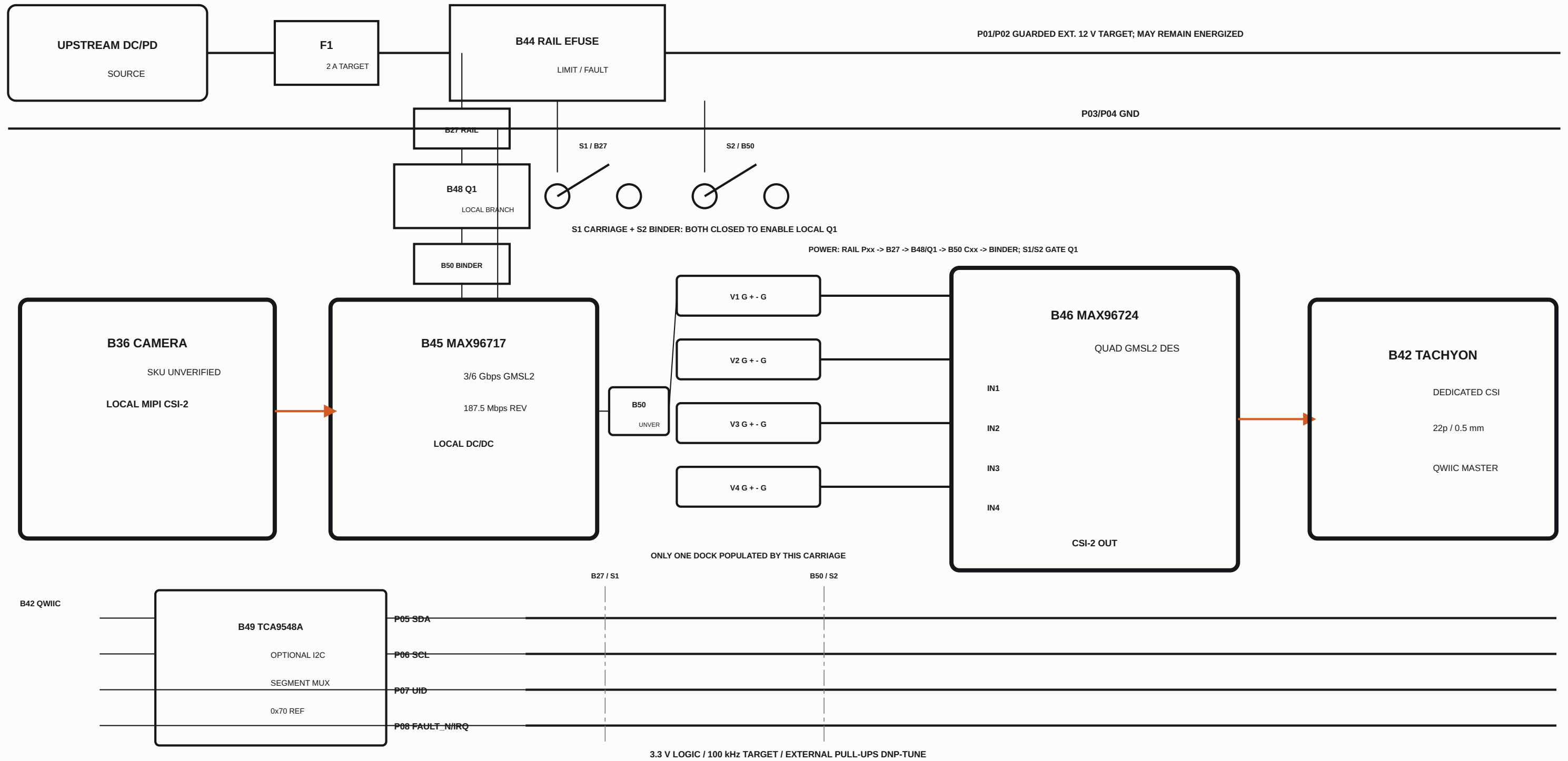
Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

NTS - DO NOT SCALE



SCALE NTS	DATE 2026-08-20	REV B-41
SHEET 08 OF 12 / S07	STATUS THEORETICAL	DRAWN GBG#41
DWG MT-S07	FORMAT A3 SVG/PDF	TOL AS NOTED

RAIL POWER / CONTROL / GMSL2 - FUNCTIONAL DIAGRAM, NOT RELEASE SCHEMATIC



MECHANISM NOTE - EE-24 INTERLOCK

EE-24 nodes: VIN_RAIL -> F1 -> VIN_FUSED -> P01||P02 VIN_SHARED.
INTERLOCK_OK = S1 NO series S2 NO. Q1_EN never driven by P08.
S1 open: Q1 off, discharge, ISO_EN off, then B27 may lift.

FAULT POLICY / NOT A RELEASE SCHEMATIC

KiCad on #24/#25 is circuit authority; this SVG is a review diagram.
S1/S2/Q1/B27/B50 series UNVERIFIED (#23). No MPNs invented.
P08 diagnostic only. GMSL P10/P11 are not MIPI. EE-26: not animal-use.

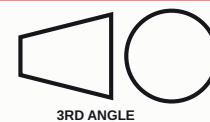
PARTICLE-BASE MANTIS TERRARIUM

ELECTRICAL + VIDEO FUNCTIONAL DIAGRAM

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

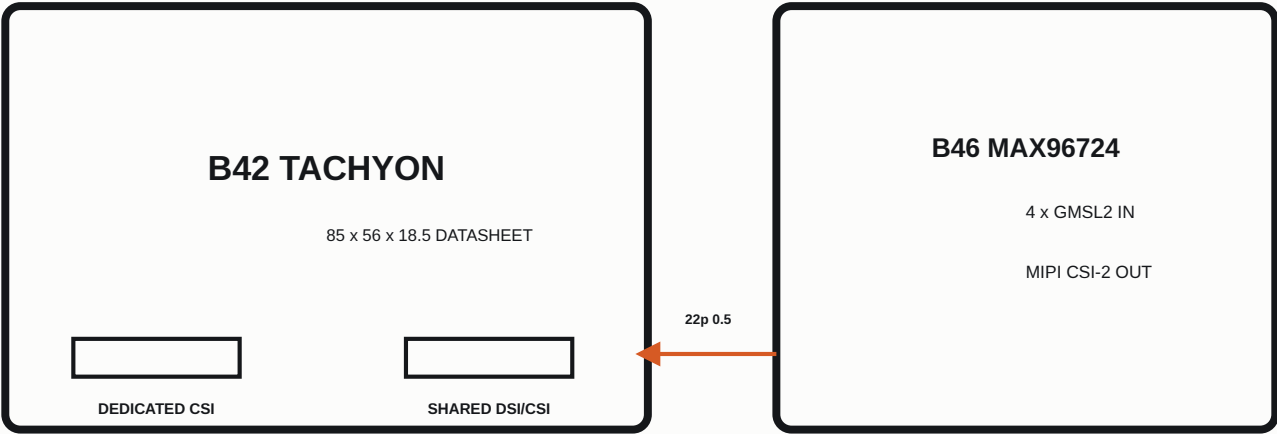
NTS - DO NOT SCALE



SCALE NTS	DATE 2026-08-20	REV B-41
SHEET 09 OF 12 / S08	STATUS THEORETICAL	DRAWN GBG#41
DWG MT-S08	FORMAT A3 SVG/PDF	TOL AS NOTED

TACHYON + DESERIALIZER CARRIER

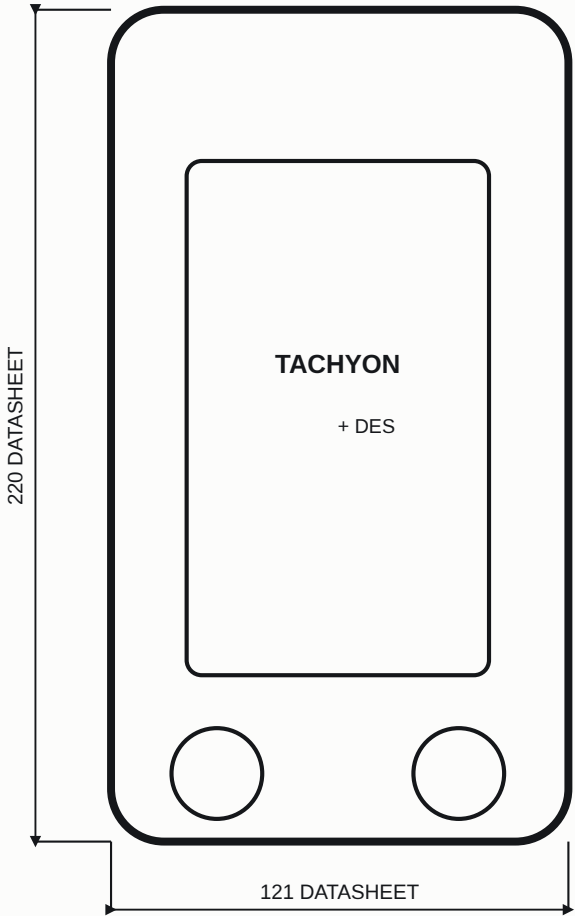
SCALE NTS - DO NOT SCALE



DESERIALIZER CARRIER DIMS / TACHYON DRIVER SUPPORT UNVERIFIED

OPTIONAL M1 EXTERNAL BRICK

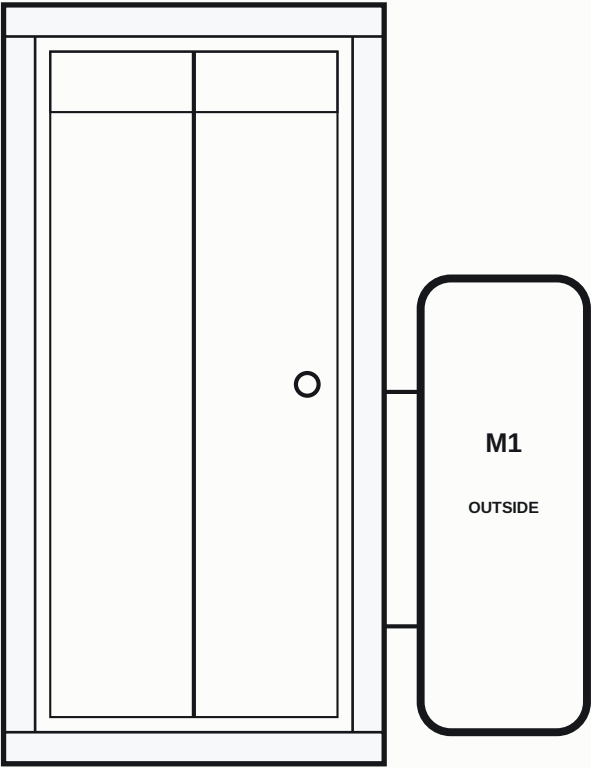
SCALE NTS - DO NOT SCALE



M20 NOT USED FOR MOVING CSI: NO CAMERA TETHER

EXTERNAL MOUNT BOUNDARY

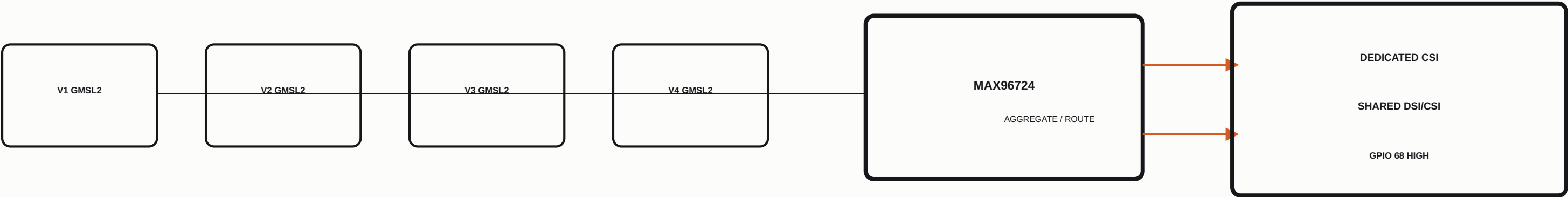
SCALE NTS - DO NOT SCALE



BRICK + SERDES HUB NEVER ENTER WET VOLUME

FOUR VIDEO DOCKS / TWO TACHYON CSI OPTIONS

SCALE NTS - DO NOT SCALE



VERIFIED THIS RUN (PARTICLE / ADI PAGES)

Tachyon 85x56x18.5; CSI 22p / 0.5 mm; 2.5 Gbps/lane.
GPIO 68 HIGH selects CSI on shared DSI/CSI2. No CSI pin table reprinted.

NOT DRAWN / UNVERIFIED

No HAT pin numbers. EE-23 transcribed CSI photos stay in the library.
IMX519/S5K3P9SX are listed sensors; orderable module SKU is not selected.

PARTICLE-BASE MANTIS TERRARIUM

SERDES / CSI / PARTICLE BRICK

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

NTS - DO NOT SCALE



SCALE
NTS

SHEET
10 OF 12 / S09

DWG
MT-S09

DATE
2026-08-20

STATUS
THEORETICAL

FORMAT
A3 SVG/PDF

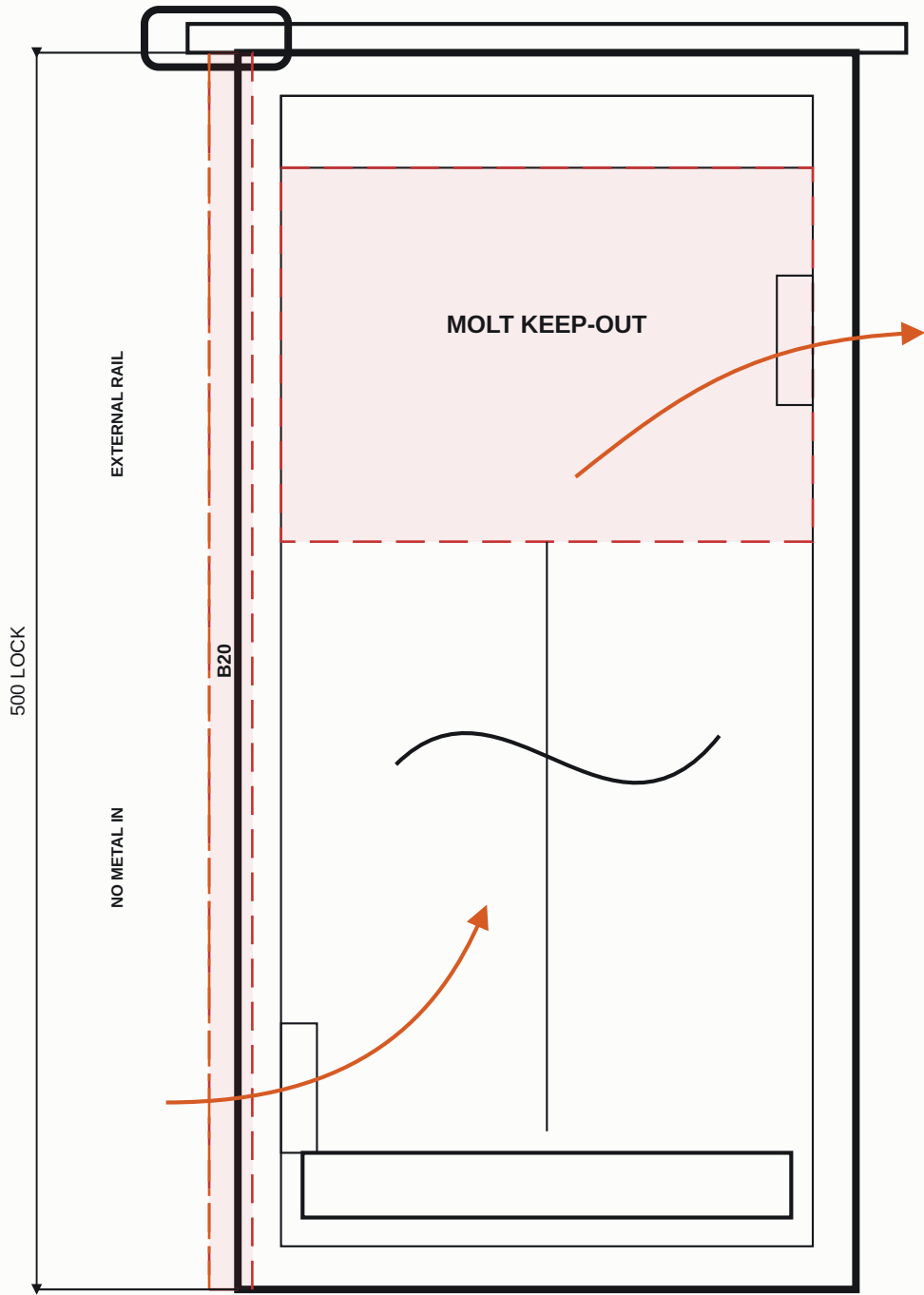
REV
B-41

DRAWN
GBG#41

TOL
AS NOTED

SECTION THROUGH ANIMAL VOLUME

SCALE NTS - DO NOT SCALE



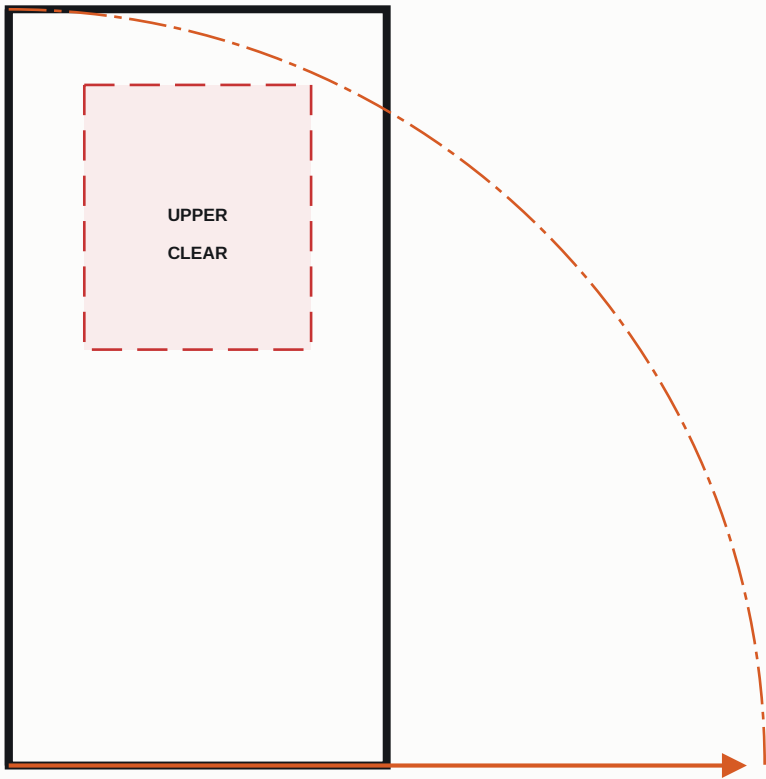
FALSE BOTTOM + 25 TRAY REF

LOW INTAKE -> HIGH EXHAUST

NOMINAL CLEAR 202 W x 202 D x 427 H CALCULATED

DOOR SWING / SERVICE ENVELOPE

SCALE NTS - DO NOT SCALE



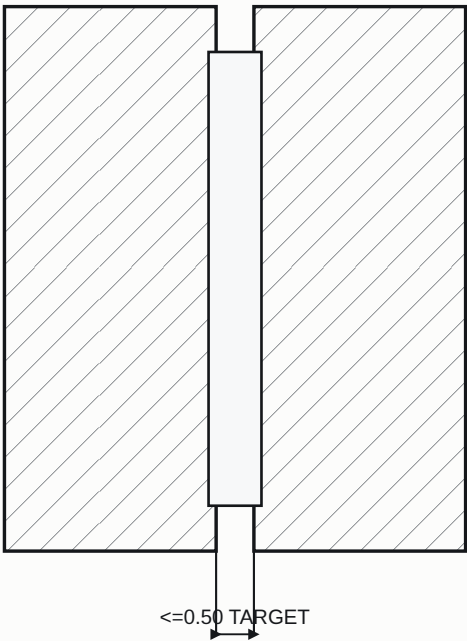
90 deg MIN SERVICE SWING

B20 KEEP-OUT (CAD-01)

Barrier bbox -8..250 X,
-8..250 Y, 0..500 Z REF.
Fused across front+left joints/splices/door.
Not a film over contacts.
Molt keep-out 142 CALCULATED.

DETAIL J - NYMPH-PROOF GAP

SCALE NTS - DO NOT SCALE



LABYRINTH + COMPRESSIBLE WIPER

SCREEN APERTURE <=0.8 LOCK

WET-VOLUME RULE

No metal mesh / copper / pogo.
PETG or ASA wet parts.
Nymph gap <=0.50 TARGET.
Solo animal. No taxon inferred.
Heat-mat pass only; no mains.

PARTICLE-BASE MANTIS TERRARIUM

HUSBANDRY INTERFERENCE

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

NTS - DO NOT SCALE



SCALE
NTS

SHEET
11 OF 12 / S10

DWG
MT-S10

DATE
2026-08-20

STATUS
THEORETICAL

FORMAT
A3 SVG/PDF

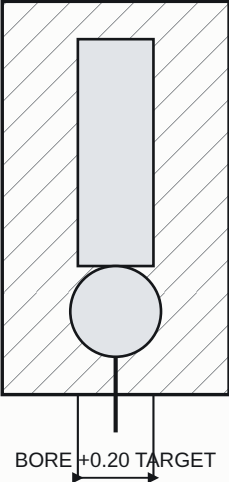
REV
B-41

DRAWN
GBG#41

TOL
AS NOTED

K - POGO BORE / TRAVEL

NTS - DO NOT SCALE



0.60 WORK / 1.20 LIFT TARGET

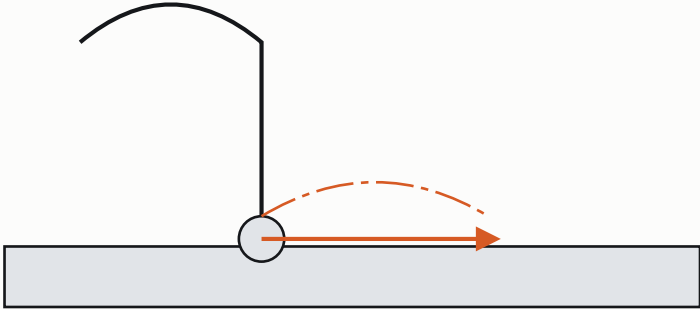
THROW 2.2 / NEED 1.8

POGO SERIES UNVERIFIED #23

BORE +0.20 TARGET

L - STRIP WIPE

NTS - DO NOT SCALE

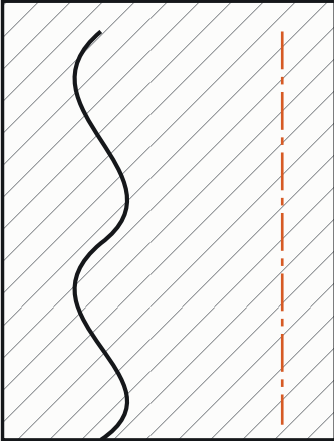


P01-P08 WIPE 0.8 TARGET

P09-P12 VERTICAL SEAT; NO WIPE

M - PINCH SPRING POCKET

NTS - DO NOT SCALE



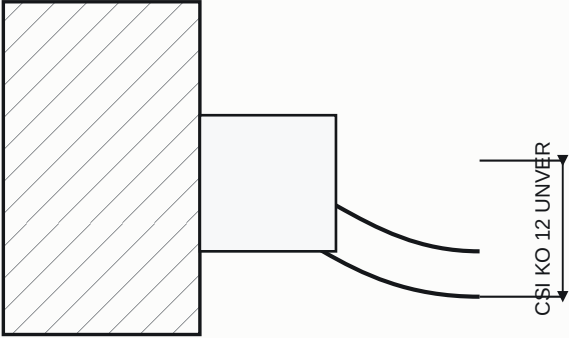
POCKET +0.30

NO SHARP ROOT

10k CYCLE TARGET

N - INTERNAL CSI / SERDES RELIEF

NTS - DO NOT SCALE

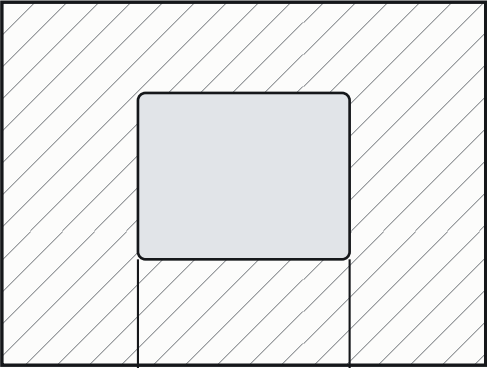


CLAMP FPC INSULATION, NEVER CONDUCTORS

CSI KO 12 UNVER

P - MAGNET POCKET

NTS - DO NOT SCALE

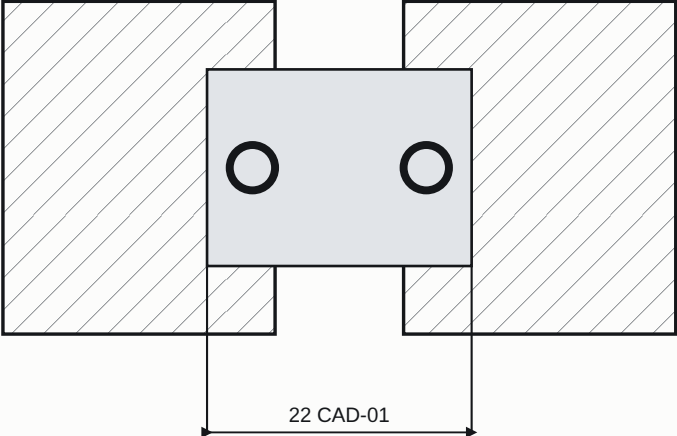


MAGNET SIZE UNVERIFIED

CAPTURE LIP; MAGNET DOES NOT DEFINE NYMPH SEAL

Q - 250/500 JOINT

NTS - DO NOT SCALE



SHEAR KEY + TWO M3; FACE CASSETTE NOT STRUCTURAL

22 CAD-01

DETAIL RELEASE RULE

TARGET/REF/UNVERIFIED values need coupon or vendor drawing before STEP release. B20 keep-out on S10 is dimensional from CAD-01 bbox, not a shop measurement.
First articles: pogo bore, strip wipe, latch pull-off, nymph-gap. Base cdd523c55a63. CAD-01 fe8f875a80b3. EE-23..26 PRs 37-40 read-only.

PARTICLE-BASE MANTIS TERRARIUM

DETAIL BLOW-UPS

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

NTS - DO NOT SCALE



SCALE NTS	DATE 2026-08-20	REV B-41
SHEET 12 OF 12 / S11	STATUS THEORETICAL	DRAWN GBG#41
DWG MT-S11	FORMAT A3 SVG/PDF	TOL AS NOTED